STAT 709: Homework 6

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All exercises are required unless marked optional.

1. Modes of convergence

Jun Shao, *Mathematical Statistics*, second edition, Chapter 1:

- Required: Exercises 127, 135, and 140.
- Optional: Exercises 137 and 152.

Clarification for Exercise 137: Use the stated convergence along every sequence of conditioning values, with conditional-law versions chosen on full-measure sets. Convergence only at each fixed conditioning value is not the assumption.

2. Asymptotic distributions and estimation

Bickel and Doksum, *Mathematical Statistics: Basic Ideas and Selected Topics*, Volume I, second edition, Chapter 5, Section 5.6, Problems for Section 5.3: Exercises 13 and 33 (required).

Correction to Exercise 33(b): The inconsistent estimator in the final sentence is the variance MLE $\widehat{\sigma}^2$. Use $k \geq 2$ observations per group, with k fixed as $p \to \infty$.